# RES-STR — Materials, Structural Response and Damage

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RES-STR records how hydrodynamic pressure, shear, force, impulse and moment histories from the local fixed-rigid barrier simulations will be transferred into later structural analyses. The active Studentship baseline is not a deformable-structure, material-damage, scour or two-way FSI model. Those topics remain stretch extensions after hydrodynamic load extraction, data provenance and the V1–V2–V3 validation hierarchy are stable.

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riers, scour or two-way fluid–structure coupling in the baseline tsunami-barrier comparison.

## 5.2 Research Questions

1. Which pressure, shear, force, impulse and moment outputs are required from the local fluid solver?
2. How are distributed fluid tractions reduced or mapped without losing resultant force and moment?
3. Which structural quantities are stretch consumers rather than active baseline outputs?

## 5.3 Terminology and Definitions

## 5.4 Literature Evidence

Baragamage and Wu [1] demonstrates the sequence from three-dimensional tsunami fluid loading to structural motion and feedback. Source role: supports preserving distributed load histories for later FSI-capable analysis. Limitation: its reduced bridge structure does not justify activating full deformable barrier analysis in the current baseline.

Istrati and Buckle [2] shows that pressure distribution can alter component demand and moment even where total force is known. Source role: supports retaining spatial pressure/traction fields, not only resultant force. Project conclusion: the local hydrodynamic model shall export complete load histories before any structural extension is attempted.

## 5.5 Governing Relations and Quantitative Definitions

## 5.6 Material or Structural System

## 5.7 Load Cases

## 5.8 Hydrodynamic Traction Transfer

The local three-dimensional fluid model supplies a transient traction field on the wetted structural surface. The traction exerted by the fluid on the solid is

$$\boldsymbol{t}_{f\to s} = \boldsymbol{\sigma}_f \boldsymbol{n}_s \tag{5.1}$$

where $\boldsymbol{n}_s$ is the outward unit normal of the current structural surface and the fluid Cauchy stress is

$$\boldsymbol{\sigma}_f = -p\boldsymbol{I} + \boldsymbol{\tau}_f \tag{5.2}$$

For a finite-element structural discretisation, the distributed interface traction is converted into the structural load vector through

$$\boldsymbol{f}_{\text{hyd}}(t) = \int_{\Gamma_{fs}(t)} \boldsymbol{N}_s^\top \boldsymbol{t}_{f\to s}(t)\,\mathrm{d}\Gamma \tag{5.3}$$

The integration is performed over the current wetted interface $\Gamma_{fs}(t)$. The transferred field must preserve the resultant force and moment and should preserve interface work when the fluid and structural surface discretisations do not match.

The tsunami FSI model of Baragamage and Wu demonstrates the sequence from three-dimensional fluid loading to structural motion and subsequent feedback to the fluid solver, although its structural representation is reduced rather than a full three-dimensional deformable continuum [1]

**Selected approach.** Hydrodynamic loading is represented as a distributed pressure and viscous traction field on the fixed barrier surface during the active baseline. Reduction to a single resultant force is not sufficient for later structural or damage studies. Decision role: Equation (5.1)–Equation (5.3) define the load-transfer contract. Limitation: the equations are a handoff requirement and do not claim that a structural solver or FSI run has been executed.

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## 5.17 Project-Specific Conclusions

The Studentship shall first stabilise hydrodynamic load extraction: pressure, shear, force, moment and impulse histories on fixed barrier patches. Material deformation and damage remain later consumers of these histories.

## 5.18 Selected Approach and Rationale

The selected near-term approach is one-way hydrodynamic load export from the local URANS–VOF model to structured output. One-way CFD-to-FEA and two-way FSI are retained as stretch paths, not baseline deliverables.

## 5.19 Unresolved Decisions

Open decisions are load-mapping tolerances, time-history sampling rate, structural mesh target, sign convention for moments and the activation criteria for one-way structural analysis.

## 5.20 Requirements and Handoffs

Software shall emit `patch_id`-resolved pressure, shear, resultant force, impulse, overturning moment, torsional moment, centre-of-pressure and integration-window metadata. RES-VER-MET shall define tolerances for force/moment consistency before RES-STR-SIM or RES-STR-FSI consumes the loads.

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# 6 RES-STR-RSP — Structural Response and Stability

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## 6.2 Research Questions

## 6.3 Terminology and Definitions

## 6.4 Literature Evidence

## 6.5 Governing Relations and Quantitative Definitions

## 6.6 Three-Dimensional Structural Equation of Motion

For the active hydrodynamic baseline, the barrier is represented in the fluid model as a fixed rigid solid. The deformable three-dimensional solid description below is retained as a future structural-response extension after hydrodynamic load histories have been validated.

Following spatial discretisation, the structural equation of motion is

$$\boldsymbol{M}_s\ddot{\boldsymbol{q}} + \boldsymbol{C}_s\dot{\boldsymbol{q}} + \boldsymbol{f}_{\text{int}}(\boldsymbol{q}, \boldsymbol{z}) = \boldsymbol{f}_{\text{hyd}}(t) + \boldsymbol{f}_{\text{body}}(t) \tag{6.1}$$

where $\boldsymbol{q}$ contains the structural displacement degrees of freedom, $\boldsymbol{z}$ contains material-history variables, $\boldsymbol{M}_s$ is the mass matrix, $\boldsymbol{C}_s$ is the damping matrix and $\boldsymbol{f}_{\text{int}}$ is the nonlinear internal-force vector.

The fixed-base boundary condition is

$$\boldsymbol{d} = \boldsymbol{0} \qquad \text{on } \Gamma_{\text{base}} \tag{6.2}$$

A nonlinear and irregular tsunami load does not itself require nonlinear material behaviour. Structural nonlinearity arises from finite deformation, nonlinear constitutive response, contact, damage or failure. These mechanisms are deferred until a separate structural activation gate is passed.

## 6.7 Updated-Lagrangian Reference Description

The Lagrangian formulation governs how the deforming solid is referenced during one tsunami simulation. In an Updated-Lagrangian formulation, the latest converged configuration becomes the reference for the following increment

$$\boldsymbol{x}_{n+1} = \boldsymbol{x}_n + \Delta \boldsymbol{d}_{n+1} \tag{6.3}$$

This differs from the barrier-class comparison loop. Each future structural case would begin with its own undeformed design geometry, while the Updated-Lagrangian formulation governs deformation within that structural simulation.

If structural response is activated, candidate geometries may undergo substantial rotation, yielding and permanent shape change. The Updated-Lagrangian description is therefore retained as the preferred finite-deformation reference formulation. It is compatible with three-dimensional transient solid dynamics and incremental constitutive updates [3]

**Deferred selected approach.** Use an Updated-Lagrangian three-dimensional finite-deformation solid model only after the fixed-rigid hydrodynamic load extraction and validation gate is passed.

### 6.8 Material or Structural System

### 6.9 Load Cases

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## 7 RES-STR-DMG — Damage, Failure and Performance-Loss Models

### 7.1 Purpose and Scope

This Deliverable records damage and failure modelling as a stretch extension. The active project baseline is hydrodynamic comparison of fixed rigid barriers; it shall not infer material damage, scour, fracture or post-failure debris motion from fluid loads alone.

## 7.2 Research Questions

## 7.3 Terminology and Definitions

## 7.4 Literature Evidence

## 7.5 Governing Relations and Quantitative Definitions

## 7.6 Material or Structural System

## 7.7 Load Cases

## 7.8 Constitutive Behaviour

## 7.9 Response and Stability Measures

## 7.10 Damage and Failure Modes

## 7.11 Initial Failure-Modelling Scope

The Updated-Lagrangian formulation permits large deformation but does not independently provide plasticity, damage, fracture or topology change. Earlier structural notes identified this as a possible future route; it is not the active baseline.

Full post-fracture continuation would additionally require crack or damage evolution, contact between separated surfaces, topology change and a fluid treatment for detached structural fragments. These capabilities are outside the active fixed-rigid hydrodynamic baseline.

**Selected scope.** Defer large nonlinear deformation, material damage, fragmentation, debris–fluid interaction and scour until the fixed-rigid hydrodynamic load histories have been validated and accepted. Source role: Baragamage and Wu [1] demonstrates that tsunami loading can be coupled to structural motion, but also supports treating such coupling as a higher-fidelity extension. Limitation: no project damage model has been selected or executed.

## 7.12 Fluid–Structure Coupling Requirements

## 7.13 Candidate Approaches

## 7.14 Critical Comparison

## 7.15 Assumptions and Applicability

## 7.16 Limitations and Failure Conditions

## 7.17 Project-Specific Conclusions

The current Studentship output is a load dataset suitable for later damage studies, not a damage prediction. Damage claims require a separate constitutive model, validation evidence and acceptance gate.

## 7.18 Selected Approach and Rationale

The selected approach is formal deferral. Damage metrics may be listed as future consumers of pressure, shear, force, impulse and moment histories, but shall not be reported as baseline simulation outputs.

## 7.19 Unresolved Decisions

Open decisions are candidate material families, damage variables, failure criteria, scour coupling and validation data for damage or post-failure behaviour.

## 7.20 Requirements and Handoffs

Software shall not implement damage or deformable-structure logic from this deliverable. It shall preserve hydrodynamic histories and geometry/patch identifiers so a later structural extension can consume them.

## 7.21 Deliverable Completion Record

# 8 RES-STR-FSI — Fluid–Structure Interaction and Coupling Fidelity

## 8.1 Purpose and Scope

This Deliverable records a future FSI capability path and the load history information needed to support it. The active Studentship baseline remains fixed terrain and fixed rigid barriers in the local URANS–VOF model. One-way CFD-to-FEA and two-way FSI are stretch extensions after hydrodynamic load extraction is stable.

## 8.2 Research Questions

## 8.3 Terminology and Definitions

## 8.4 Literature Evidence

### 8.4.1 Baragamage and Wu (2024)

Baragamage and Wu [1] demonstrated a tsunami-impact framework in which an incompressible free-surface Navier–Stokes model was coupled with a structural equation of motion. Fluid pressure generated structural force, while structural displacement and velocity were returned to the fluid boundary.

The study establishes the feasibility of two-way tsunami FSI but does not establish that two-way coupling is required for every barrier candidate. Its bridge representation was structurally reduced, and parts of the validation used a constrained flow configuration.

### 8.4.2 Istrati and Buckle (2019)

Istrati and Buckle [2] demonstrated that the distribution of pressure and structural demand cannot always be reconstructed from the total force alone. Changes in internal ventilation altered uplift, moment, and the distribution of demand between structural components.

## 8.5 Governing Relations and Quantitative Definitions

### 8.5.1 Structural State and Momentum Equation

The structural state is provisionally defined as:

$$\mathcal{Q}_\mathrm{s} = \left\{\mathbf{d}_\mathrm{s}, \dot{\mathbf{d}}_\mathrm{s}, \boldsymbol{\sigma}_\mathrm{s}, \boldsymbol{\xi}_\mathrm{s}\right\} \tag{8.1}$$

where $\boldsymbol{\xi}_\mathrm{s}$ contains the internal variables required by the selected material model.

In continuum form, structural momentum conservation is:

$$\rho_\mathrm{s}\ddot{\mathbf{d}}_\mathrm{s} = \nabla_0 \cdot \mathbf{P}_\mathrm{s} + \rho_\mathrm{s}\mathbf{b}_\mathrm{s} \tag{8.2}$$

where $\mathbf{P}_\mathrm{s}$ is the first Piola–Kirchhoff stress tensor.

The corresponding finite-element semi-discrete form is:

$$\mathbf{M}_s\ddot{\mathbf{d}}_s + \mathbf{C}_s\dot{\mathbf{d}}_s + \mathbf{f}_{\text{int}}(\mathbf{d}_s, \boldsymbol{\xi}_s) = \mathbf{f}_{\text{fluid}} + \mathbf{f}_{\text{ext}} \tag{8.3}$$

The formulation shall support more than a single structural degree of freedom. The initial structural model may be linear elastic, but the interface shall permit later activation of:

- geometric nonlinearity;
- material plasticity;
- contact and connection behaviour;
- damage and failure variables;
- rigid-body translation and rotation.

## 8.6 Fluid–Structure Interaction Model and Activation Strategy

The structural formulation shall therefore preserve the spatial and temporal hydrodynamic traction even where the optimisation initially uses a fixed rigid barrier.

For the current project scope, this statement is a data-preservation requirement only: fixed-barrier simulations must retain enough surface history for later structural studies, but they do not move the barrier or solve structural response.

## 8.7 Material or Structural System

## 8.8 Load Cases

## 8.9 Constitutive Behaviour

## 8.10 Response and Stability Measures

## 8.11 Damage and Failure Modes

## 8.12 Fluid–Structure Coupling Requirements

### 8.12.1 Fluid–Structure Interface Conditions

At the fluid–structure interface $\Gamma_{\text{fs}}$, kinematic compatibility requires:

$$\mathbf{u}_f = \dot{\mathbf{d}}_s \qquad \text{on } \Gamma_{\text{fs}} \tag{8.4}$$

Dynamic equilibrium requires:

$$\boldsymbol{\sigma}_{\mathrm{f}}\mathbf{n}_{\mathrm{f}} + \boldsymbol{\sigma}_{\mathrm{s}}\mathbf{n}_{\mathrm{s}} = \mathbf{0} \qquad \text{on } \Gamma_{\mathrm{fs}} \tag{8.5}$$

The structural load is obtained from the mapped fluid traction:

$$\mathbf{f}_{\mathrm{fluid}} = \mathcal{T}_{\mathrm{f}\to\mathrm{s}}\left[\boldsymbol{\sigma}_{\mathrm{f}}\mathbf{n}_{\mathrm{f}}\right] \tag{8.6}$$

where $\mathcal{T}_{\mathrm{f}\to\mathrm{s}}$ is the fluid-to-structure load-transfer operator.

For two-way coupling, the deformed structural state is returned through:

$$\Gamma_{\mathrm{fs}}^{n+1} = \mathcal{T}_{\mathrm{s}\to\mathrm{f}}\left[\mathbf{d}_{\mathrm{s}}^{n+1}\right] \tag{8.7}$$

The transfer operators shall conserve resultant force and moment to within the tolerances defined by ‘RES-VER-MET‘.

### 8.12.2 FSI Fidelity Hierarchy

The structural-coupling hierarchy is:

| Mode | Fluid treatment | Structural treatment | Intended use |
|---|---|---|---|
| 0 | Local URANS | Fixed rigid barrier | Initial fixed-rigid hydrodynamic comparison |
| 1 | Local URANS or LES | One-way CFD-to-FEA | Stress, deformation, stability, and failure screening |
| 2 | Local URANS or LES | Partitioned two-way FSI | Cases with material but moderate deformation feedback |
| 3 | Local high-fidelity model | Strongly coupled nonlinear FSI | Flexible, moving, unstable, or intentionally compliant final designs |

**Table 1.** Fluid–structure coupling hierarchy for candidate and shortlisted barrier designs

### 8.12.3 Two-Way FSI Activation Gate

Two-way FSI shall be activated where one or more of the following conditions are satisfied:

1. structural displacement changes the exposed geometry or wetted area materially;
2. structural motion changes a flow passage, opening, or overtopping path;
3. structural velocity is no longer negligible relative to the adjacent fluid velocity;

4. rotation or translation changes the reflected or transmitted wave field;
5. connection flexibility controls the global response;
6. instability, uplift, sliding, overturning, or large rotation is predicted;
7. the barrier relies on intentional compliant motion for energy dissipation;
8. one-way and two-way pilot simulations produce materially different hydrodynamic or structural performance measures.

A preliminary motion-feedback indicator may be defined as:

$$\Pi_{\mathrm{fsi}} = \max_{\Gamma_{\mathrm{fs}},t} \left( \frac{\|\dot{\mathbf{d}}_{\mathrm{s}}\|}{\|\mathbf{u}_{\mathrm{f}}\| + \epsilon_{\mathrm{u}}} \right) \tag{8.8}$$

where $\epsilon_{\mathrm{u}}$ prevents division by zero.

The threshold for activating two-way FSI shall be determined through sensitivity testing rather than assigned universally.

## 8.13 Reference Descriptions of the Coupled Domains

The regional and local fluid models use an Eulerian description in which fluid variables are evaluated at spatial positions

$$\boldsymbol{u} = \boldsymbol{u}(\boldsymbol{x}, t) \tag{8.9}$$

The solid uses a Lagrangian material description

$$\boldsymbol{x} = \boldsymbol{\chi}(\boldsymbol{X}, t) = \boldsymbol{X} + \boldsymbol{d}(\boldsymbol{X}, t) \tag{8.10}$$

Impact does not require a Lagrangian fluid description. Retaining an Eulerian fluid mesh is more suitable for breaking, overtopping and free-surface topology change, while the Lagrangian solid description tracks the barrier deformation and material history.

## 8.14 Fluid–Solid Interface Conditions

Two-way coupling requires kinematic compatibility and dynamic equilibrium on the moving interface

$$\boldsymbol{u}_f = \dot{\boldsymbol{d}}_s \qquad \text{on } \Gamma_{fs}(t) \tag{8.11}$$
$$\boldsymbol{\sigma}_f \boldsymbol{n}_f + \boldsymbol{\sigma}_s \boldsymbol{n}_s = \mathbf{0} \qquad \text{on } \Gamma_{fs}(t) \tag{8.12}$$

The fluid solver transfers traction to the structure. The structural solver returns the current interface position and velocity to the fluid representation. A one-way calculation omits the return of structural motion, while a two-way calculation repeats this exchange throughout the transient solution.

## 8.15 Moving-Boundary Treatment

A body-fitted Arbitrary Lagrangian–Eulerian method moves the fluid mesh with the structure. The fluid advection velocity is then evaluated relative to the mesh velocity

$$\boldsymbol{u}_{\text{rel}} = \boldsymbol{u} - \boldsymbol{w}_m \tag{8.13}$$

ALE provides a direct conforming interface and conventional near-wall treatment. Its principal limitation for the present application is the risk of fluid-mesh distortion under severe structural deformation, followed by remeshing and transfer of pressure, velocity, VOF and turbulence fields. It remains useful as a reference method for moderate-deformation benchmarks [4]

If FSI is activated as a stretch extension, the selected reference family is a sharp immersed structural surface with conservative cut cells on a fixed Eulerian background mesh. The method avoids global fluid-mesh deformation while accounting geometrically for the changing fluid volume and moving solid boundary. Pasquariello et al. demonstrate conservative finite-volume–finite-element coupling with cut cells and nonmatching interface transfer for three-dimensional FSI with large structural deformation [5]. Baragamage and Wu apply a related immersed-boundary and cut-cell strategy to three-dimensional tsunami loading [1]

**Deferred selected approach.** Retain a sharp immersed-boundary representation with conservative cut-cell treatment as the preferred FSI extension path. Retain ALE as a comparator for moderate-deformation verification cases.

The selected family must subsequently define cut-cell geometry, changing fluid volumes, small-cell stabilisation, reconstructed wall states, traction recovery and conservative transfer between fluid and structural interfaces.

## 8.16 Deferred High-Level FSI Architecture

The deferred high-level architecture is

$$\begin{aligned}
&\text{Eulerian polyhedral finite-volume two-phase fluid}\\
&\quad + \text{VOF free-surface capture}\\
&\quad + \text{Updated-Lagrangian three-dimensional solid FEM}\\
&\quad + \text{generalised-}\alpha\text{ structural time integration}\\
&\quad + \text{Newton-type nonlinear structural solution}\\
&\quad + \text{sharp immersed boundary with conservative cut cells}
\end{aligned} \tag{8.14}$$

The following high-level decisions remain unresolved:

1. monolithic or partitioned coupling
2. loose or strong partitioned coupling if the partitioned family is selected
3. added-mass stabilisation, relaxation and interface quasi-Newton acceleration
4. conservative transfer of force, moment, displacement and interface work
5. coordination of fluid and structural time levels
6. transition criteria between rigid, one-way and two-way FSI calculations

## 8.17 Candidate Approaches

## 8.18 Critical Comparison

## 8.19 Assumptions and Applicability

## 8.20 Limitations and Failure Conditions

## 8.21 Project-Specific Conclusions

### 8.21.1 Project-Specific FSI Decision

The software and data architecture shall be FSI-capable from the outset. The baseline hydrodynamic comparison model shall nevertheless use a fixed rigid barrier and retain complete surface traction histories.

The initial structural verification shall use one-way CFD-to-FEA load transfer. Two-way FSI shall be introduced only where the activation gate demonstrates that structural motion feeds back materially into the hydrodynamic response.

This approach retains the equation-level capability for coupled simulation without imposing fully coupled FSI on the active barrier comparison campaign.

## 8.22 Selected Approach and Rationale

## 8.23 Unresolved Decisions

## 8.24 Requirements and Handoffs

Software shall preserve load, geometry and patch histories in a form that a later FSI extension can consume, but shall not implement moving boundaries, structural time integration or two-way coupling from this Research update. RES-VER-MET shall define force/moment transfer consistency metrics before FSI activation.

### 8.25 Deliverable Completion Record

# 9 RES-STR-SIM — Structural Simulation and Finite-Element Methodology

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## 9.12 Structural Spatial and Temporal Discretisation

The deferred structural solution stack consists of three-dimensional finite elements, Updated-Lagrangian finite-deformation kinematics, generalised-$\alpha$ time integration and Newton-type nonlinear equilibrium iterations. The generalised-$\alpha$ method supplies the temporal discretisation only; it does not replace the finite-element formulation, constitutive law or nonlinear solver. This stack is not part of the active fixed-rigid hydrodynamic baseline.

At each time step, equilibrium is enforced at weighted intermediate states through a residual of the form

$$\boldsymbol{R}_s = \boldsymbol{M}_s \ddot{\boldsymbol{q}}_{n+1-\alpha_m} + \boldsymbol{C}_s \dot{\boldsymbol{q}}_{n+1-\alpha_f} + \boldsymbol{f}_{\text{int}}\big(\boldsymbol{q}_{n+1-\alpha_f}, \boldsymbol{z}_{n+1}\big) - \boldsymbol{f}_{\text{ext},n+1-\alpha_f} \tag{9.1}$$

The weighted acceleration and displacement states are

$$\ddot{\boldsymbol{q}}_{n+1-\alpha_m} = (1-\alpha_m)\,\ddot{\boldsymbol{q}}_{n+1} + \alpha_m \ddot{\boldsymbol{q}}_n \tag{9.2}$$

$$\boldsymbol{q}_{n+1-\alpha_f} = (1-\alpha_f)\,\boldsymbol{q}_{n+1} + \alpha_f \boldsymbol{q}_n \tag{9.3}$$

The displacement and velocity are updated through Newmark-type relations

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + \Delta t \dot{\boldsymbol{q}}_n + \Delta t^2 \left[ \left( \frac{1}{2} - \beta \right) \ddot{\boldsymbol{q}}_n + \beta \ddot{\boldsymbol{q}}_{n+1} \right] \tag{9.4}$$

$$\dot{\boldsymbol{q}}_{n+1} = \dot{\boldsymbol{q}}_n + \Delta t \left[ (1-\gamma)\,\ddot{\boldsymbol{q}}_n + \gamma \ddot{\boldsymbol{q}}_{n+1} \right] \tag{9.5}$$

Under the standard parameter conditions, the method provides second-order accuracy and controllable high-frequency numerical dissipation [6]. Liu and Marsden demonstrate its use within a three-dimensional nonlinear solid and FSI framework [7]

## 9.13 Nonlinear Equilibrium Solution

If structural response is activated, finite deformation and nonlinear constitutive behaviour require an iterative solution at each structural time step. A Newton correction is obtained from

$$\boldsymbol{K}_{\text{eff}}^{(k)} \Delta \boldsymbol{q}^{(k)} = -\boldsymbol{R}_s^{(k)} \tag{9.6}$$

$$\boldsymbol{q}^{(k+1)} = \boldsymbol{q}^{(k)} + \Delta \boldsymbol{q}^{(k)} \tag{9.7}$$

where $\boldsymbol{K}_{\text{eff}}$ contains the mass, damping and consistent tangent-stiffness contributions. The structural state is accepted only after the residual and displacement correction satisfy the selected nonlinear convergence criteria.

**Deferred selected approach.** Use generalised-$\alpha$ time integration embedded in a Newton-type nonlinear finite-element solution only after hydrodynamic load extraction is accepted.

**Unresolved decisions.** The constitutive family, element technology, incompressibility treatment, damping model, spectral-radius parameter and nonlinear convergence tolerances remain to be selected.

## 9.14 Candidate Approaches

## 9.15 Critical Comparison

## 9.16 Assumptions and Applicability

## 9.17 Limitations and Failure Conditions

## 9.18 Project-Specific Conclusions

## 9.19 Selected Approach and Rationale

## 9.20 Unresolved Decisions

## 9.21 Requirements and Handoffs

## 9.22 Deliverable Completion Record

# 10 RES-STR-SPEC — Integrated Structural and Damage Specification

## 10.1 Purpose and Scope

This Deliverable consolidates the current structural decision: baseline simulations treat barriers as fixed rigid geometry and export hydrodynamic pressure, shear, force, impulse and moment histories. Material response, deformation, damage, scour and two-way FSI are stretch extensions requiring separate evidence and validation.

## 10.2 Research Questions

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## 10.4 Literature Evidence

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## 10.11 Fluid–Structure Coupling Requirements

## 10.12 Candidate Approaches

## 10.13 Critical Comparison

## 10.14 Assumptions and Applicability

## 10.15 Limitations and Failure Conditions

## 10.16 Project-Specific Conclusions

The integrated structural output for the current Studentship is a load-transfer specification, not a structural-damage model.

## 10.17 Selected Approach and Rationale

The selected active approach is fixed-rigid hydrodynamic load extraction with structured one-way handoff. Deformable structural simulation and two-way FSI remain deferred.

## 10.18 Unresolved Decisions

## 10.19 Requirements and Handoffs

Software shall preserve load histories, geometry patch identifiers and integration metadata; it shall not implement damage, structural FEM or moving-boundary FSI as part of this Research handoff.

## 10.20 Deliverable Completion Record

# References


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